

AUTOMOBILE DASHBOARD ANALYSIS

Introduction

The dataset includes various columns that provide detailed information about car sales, including financial aspects (sale price, cost price, discounts), logistical aspects (delivery charges, spare parts, labour costs), client details, and vehicle details (make, model, colour, type).

This comprehensive data allows for in-depth analysis of sales performance, cost management, and customer preferences across different regions and time periods.

Objective:

The primary objective of the analysis is to gain insights into the sales performance of different car brands across various countries, track sales trends over time, and identify patterns and relationships in the dataset.

This can help in understanding customer preferences, optimizing inventory, and improving sales strategies.

Methodology:

Data Collection-Collected Automobile data containing Invoice Date, Make, Country name, sale price, cost price, total discount, delivery charge, spare parts, labour cost, client name, model, colour, reporting year, reporting month, registration date, vehicle type, profit, total cost

Data Source: Excel

DAX:

Aston Martin Sales = CALCULATE(SUM(Automobile[SalePrice]),FILTER(Automobile,Automobile[Make]="Aston Martin"))

AverageSales = AVERAGE(Automobile[SalePrice])

Distinct_Customer = DISTINCTCOUNT(Automobile[ClientName])

MOM Growth% = ([total_saleprice]-[previous_month])/[previous_month]

Previous Quarter = CALCULATE([total_saleprice],PREVIOUSQUARTER(Calander_table[Date]))

previous_month = CALCULATE([total_saleprice],PREVIOUSMONTH(Calander_table[Date]))

QOQ Growth% = ([total_saleprice]-[Previous Quarter])/[Previous Quarter]

Rolling3Months =

CALCULATE(AVERAGE(Automobile[SalePrice]),DATESINPERIOD(Calander_table[Date],LASTDATE(Calander_table[Date]),-3,MONTH))

Year_to_Date = CALCULATE(SUM(Automobile[SalePrice]),DATESYTD(Automobile[InvoiceDate]))

Dashboard Development Designed dashboard visuals:

KPI Cards

Sales Contribution per category

Country wise sales

Total Sales

Unique customers

Last 12 months sales

Average 3 months sales

Sales YTD

Total Profit Margin

Total Cost Price

Total Sale price

Total Spare Parts

Charts Used

Stacked bar chart- Sales price by country

Clustered column chart-Sales Price and Profit by Year, profit margin by month, sale price by quarter

Funnel chart-Profit by Make

Line Charts -Sale price by Year, Sale price by Month

Scatter Plot -Sale price and profit by make and model

Pie Chart -Sales price by vehicle type

Decomposition Tree- Sale Price Drill Down>year>quarter>colour>make

Tables-Sales price ranking, year wise sales, top 5 products, monthly average sales, sales price by country and make, MOM Growth, quarter to quarter growth

Waterfall chart-Sale price by colour and country name

Tools & Technologies Used

Power BI

Desktop Power Query

DAX

Other Parameters depending upon the Projects:

Data source identification (Excel)

Data quality check (missing values, duplicates)

Data transformation & cleaning

KPI selection based on goals

Dashboard performance optimization

Visualization design & layout

Data Modelling

Requirement Analysis:

The project is based on creating an interactive automobile sales dashboard in Power BI Desktop to analyse vehicle sales, profit, customer trends, country performance, and product performance.

The dashboard should help management:

- ✓ Monitor total automobile sales and profit.
- ✓ Compare sales across countries, years, months, and quarters.
- ✓ Identify top-performing vehicle brands and models.
- ✓ Track profit margins and growth trends.
- ✓ Analyse customer and product contribution.
- ✓ Support business decision-making using visual reports.

Functional Requirements

The dashboard includes Sales analysis by:

Country

Vehicle type

Year

Month

Quarter

Brand (Make)

Profit analysis by:

Make

Year

Quarter

Month

Growth analysis:

Month-over-Month (MOM)

Quarter-over-Quarter (QOQ)

Ranking analysis for top-selling products.

Filters/Slicers for:

Model

KPI cards for:

Total Sales

YTD Sales

Last 12 Months Sales

Unique Customers

Data Requirements The dataset should contain:

Sale Price

Cost Price

Profit

Profit Margin

Vehicle Type

Country Name

Year/Month/Quarter

Make and Model

Color

Customer details

Interactive visualizations using charts, cards, slicers, and tables.

AUTOMOBILE DASHBOARD ANALYSIS

Sales Contribution Per Category

11M
Aston Marti...

5M
Bentley Sales

6M
Jaguar Sales

1M
MGB Sales

7M
Rolls Royce ...

875K
Triumph

499K
TVR

Total Sales

32M

Spain Sales

208K

Switzerland Sales

1M

Germany Sales

146K

France Sales

3M

UK Sales

16M

USA Sales

12M

Unique Customers

31

Last 12 Months Sales

\$14.96729M

Average 3 months

65K

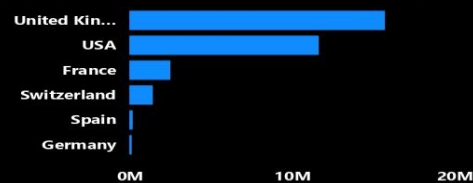
Sales YTD

\$15M

Model

- ☐ Select all
- ☐ Arnage
- ☐ Azure
- ☐ Camargue
- ☐ Cerbera
- ☐ Continental
- ☐ DB4
- ☐ DB7
- ☐ DB9
- ☐ DBS
- ☐ GT
- ☐ Phantom

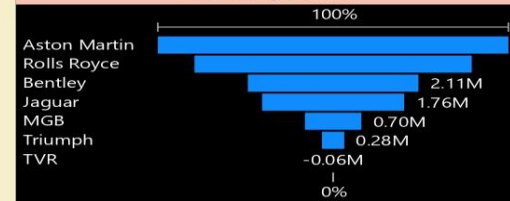
SalePrice by Country



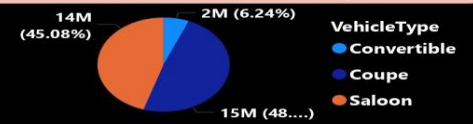
Sale Price and Profit by Year



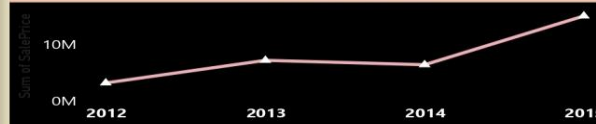
Profit by Make



Sale Price by Vehicle Type



Sum of SalePrice by Year



Sum of SalePrice by Month



Year
2012

Quarter
Qtr 1

Color
Black

Make
Jaguar



Sale Price and Sum of profit by Model and Make

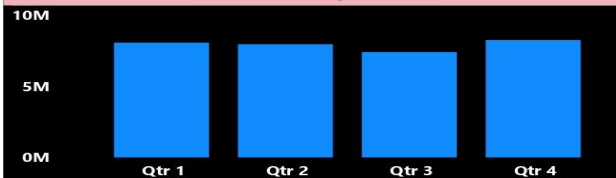


AUTOMOBILE DASHBOARD ANALYSIS

Profit Margin by Month



Sale Price by Quarter



Profit Visuals

Sale Price Visuals

MOM Growth

Month	total_saleprice	previous_month	MOM Growth%
January	2892250		Infinity
February	2522750	203500	1139.68%
March	2659300	177000	1402.43%
April	2415150	172500	1300.09%
May	2943440	246000	1096.52%
June	2603130	256500	914.87%
July	2627150	259500	912.39%
August	2308000	279000	727.24%
September	2472190	350000	606.34%
October	2561070	297500	760.86%
November	3166760	249000	1171.79%
December	2526750	333500	657.65%
Total	31697940		Infinity

Sum of ProfitMargin



Sum of CostPrice



Sum of SalePrice



Sum of SpareParts



Sale Price Ranking

Make	Sum of SalePrice	ranking
Aston Martin	10686040	1
Bentley	4951250	4
Jaguar	6319000	3
MGB	1011000	5
Rolls Royce	7356900	2
Triumph	875000	6
TVR	498750	7
Total	31697940	1

Year Wise Sales

Year	Sum of SalePrice
2012	3185500
2013	7158710
2014	6386440
2015	14967290
Total	31697940

Monthly Average Sales

Month	AverageSales
January	76,111.84
February	70,076.39
March	64,860.98
April	63,556.58
May	70,081.90
June	74,375.14
July	67,362.82
August	69,939.39
September	66,815.95
October	69,218.11
Total	69,360.92

Top 5 Products

Make	Top 5 Products
Aston Martin	10686040
Bentley	4951250
Jaguar	6319000
MGB	1011000
Rolls Royce	7356900

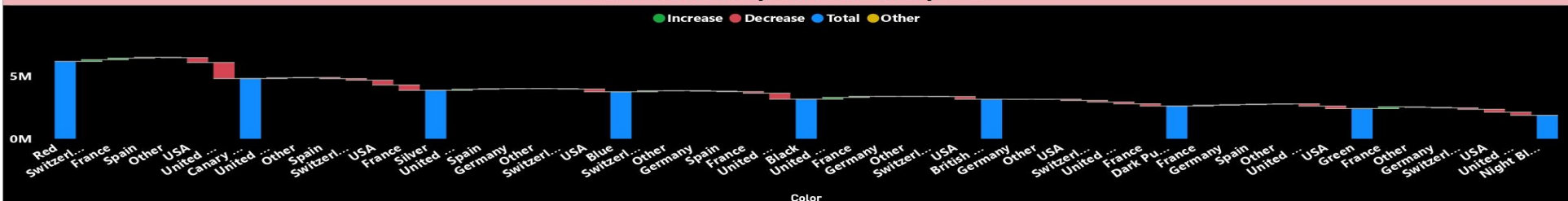
Quarter to Quarter Growth

Quarter	Sum of SalePrice	Previous Quarter	QOQ Growth%
Qtr 1	8074300		Infinity
Qtr 2	7961720	553000	1339.73%
Qtr 3	7407340	762000	872.09%
Qtr 4	8254580	926500	790.94%
Total	31697940		Infinity

SalePrice by Country & Make

CountryName	Make	Sum of SalePrice
France	Aston Martin	1487210
Switzerland	Aston Martin	543170
United Kingdom	Aston Martin	4933500
USA	Aston Martin	3722160
France	Bentley	394250
Spain	Bentley	46750
Switzerland	Bentley	222750
United Kingdom	Bentley	2407750
USA	Bentley	1879750
France	Jaguar	212000
Germany	Jaguar	74750
Spain	Jaguar	69250
Switzerland	Jaguar	450250
United Kingdom	Jaguar	2761750
USA	Jaguar	2751000
United Kingdom	MGB	663000
USA	MGB	348000
France	Rolls Royce	373300
Switzerland	Rolls Royce	192300
United Kingdom	Rolls Royce	4319750
USA	Rolls Royce	2471550
France	Triumph	28000
Spain	Triumph	47750
United Kingdom	Triumph	530250
USA	Triumph	269000
France	TVR	29750
Germany	TVR	71000
Spain	TVR	44000
Switzerland	TVR	32500
United Kingdom	TVR	109000
Total		31697940

Sum of SalePrice by Color and CountryName



Insights-

1. Total Sales Performance

Total sales reached approximately \$31.7M.

Sales YTD are around \$15M.

2. Top Performing Brands

Aston Martin generated the highest sales:

Around 10.68M sales.

Rolls-Royce ranked second with approximately 7.35M sales.

TVR had the lowest sales contribution.

3. Country-wise Sales Insights

The United Kingdom contributed the highest sales among all countries.

United States was the second-largest market.

Germany and Spain contributed comparatively lower sales.

4. Vehicle Type Analysis

Coupe and Saloon vehicle types contributed the majority of sales.

Convertible vehicles contributed the smallest share.

5. Year-wise Growth

Sales increased significantly from 2012 to 2015:

2012: 3.18M

2013: 7.15M

2014: 6.38M

2015: 14.96M

2015 recorded the highest sales performance.

6. Monthly Sales Trend

November recorded one of the highest monthly sales values.

April and August showed comparatively lower sales.

7. Profit Analysis

Most brands generated positive profits.

One brand showed a slight negative profit contribution, indicating possible high operational or production costs.

8. Customer Insights

The business served around 31 unique customers.

Conclusion

The Automobile Dashboard successfully provides a complete view of automobile sales and profitability using interactive visualizations in Power BI Desktop.

Key findings include:

- Strong growth in sales over the years.
- The United Kingdom and United States are the strongest markets.
- Aston Martin and Rolls-Royce are the top revenue-generating brands.
- Certain brands and vehicle types require improvement due to lower sales or profit margins.
- The dashboard helps management identify trends, improve decision-making, and monitor business performance effectively.
- Overall, the project demonstrates how business intelligence tools can transform raw automobile sales data into meaningful insights for strategic planning and operational improvement